



NORTHERN BEACHES SECONDARY COLLEGE

# MANLY SELECTIVE CAMPUS

Year 12

Trial Examination

2022

## Mathematics Extension II

### General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Write your name on the front of every booklet.
- In Questions 11 to 16 show relevant mathematical reasoning and/or calculations.
- NESA approved calculators and templates may be used.
- Weighting: 30%

### Section I Multiple Choice

- 10 marks
- Attempt all questions.
- Answer Sheet provided
- Allow about 15 minutes for this section

### Section II Free Response

- 90 marks
- Start a separate booklet for each question.
- Each question is of equal value.
- All necessary working should be shown in every question.
- Allow about 2 hours and 45 minutes for this section.

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

---

Q1. Given that  $1 - i$  is a solution of  $z^3 - 4z^2 + 6z - 4 = 0$ , what are the other solutions?

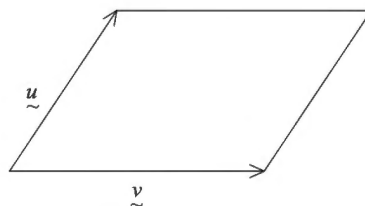
- A.  $1 + i$  and  $-6$
- B.  $1 + i$  and  $2$
- C.  $-1 - i$  and  $4 + 2i$
- D.  $1 + i$  and  $-4$

Q2. Let  $\underline{u} = 2\underline{i} - a\underline{j} - \underline{k}$  and  $\underline{v} = 3\underline{i} + 2\underline{j} - b\underline{k}$ .

If these vectors are perpendicular to each other, what are the possible values for  $a$  and  $b$ ?

- A.  $a = 2$  and  $b = 2$
- B.  $a = -2$  and  $b = 10$
- C.  $a = 0$  and  $b = 0$
- D.  $a = -1$  and  $b = -8$

Q3. The diagram below shows a rhombus.

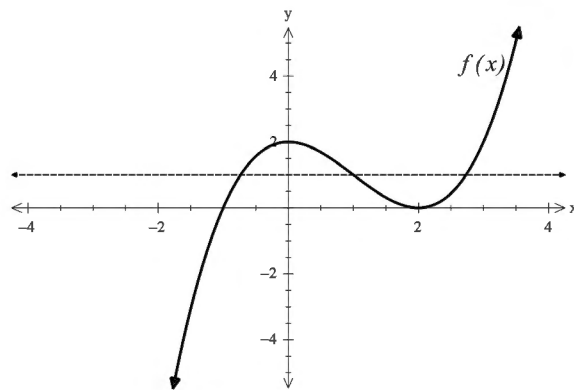


Adjacent sides are represented by two vectors,  $\underline{\underline{u}}$  and  $\underline{\underline{v}}$ .

It follows that:

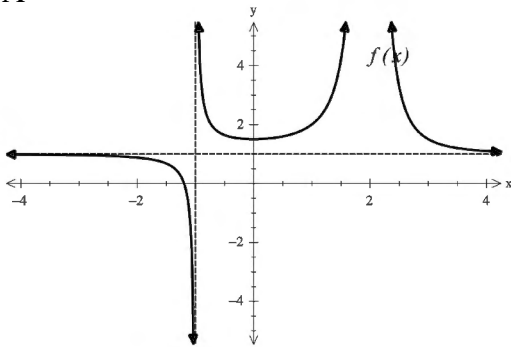
- A.  $\underline{\underline{u}} \cdot \underline{\underline{v}} = 0$
- B.  $\underline{\underline{u}} = \underline{\underline{v}}$
- C.  $(\underline{\underline{u}} + \underline{\underline{v}}) \cdot (\underline{\underline{u}} - \underline{\underline{v}}) = 0$
- D.  $|\underline{\underline{u}} + \underline{\underline{v}}| = |\underline{\underline{u}} - \underline{\underline{v}}|$

Q4. The graph of the function  $y = f(x)$  is shown below.

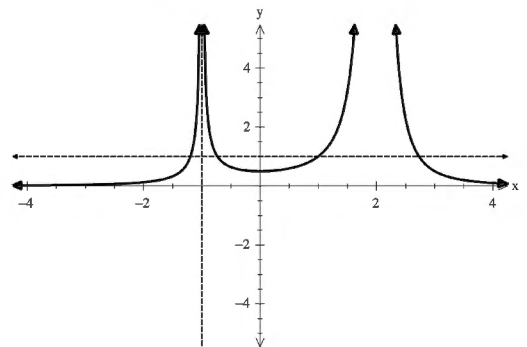


Which graph below shows the most accurate representation of the graph  $y = \frac{1}{f(x)}$ ?

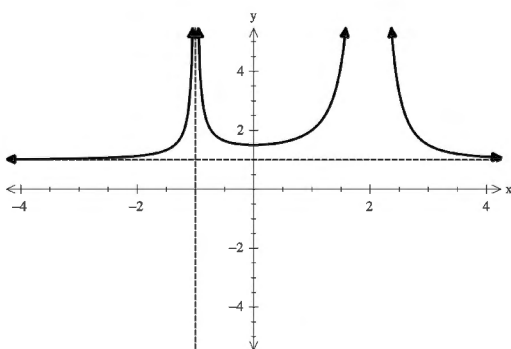
A



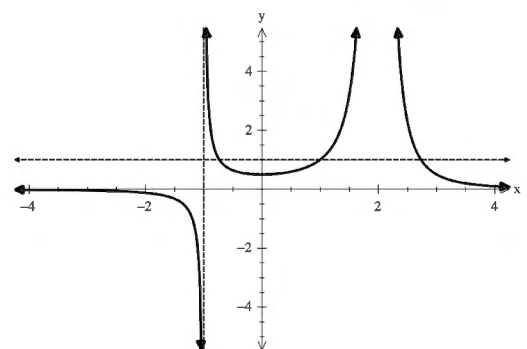
B



C



D



Q5. Consider the statement:

*If the soup contains lemon or vinegar, the soup does not contain chilli.*

Which of the following is logically equivalent to the statement?

- A. If the soup contains chilli, the soup does not contain lemon nor vinegar.
- B. If the soup contains chilli, the soup does not contain lemon or the soup does not contain vinegar.
- C. If the soup does not contain lemon or the soup does not contain vinegar, the soup contains chilli.
- D. If the soup does not contain lemon nor vinegar, the soup contains chilli.

Q6. The vector equation of the line that passes through the points A (1, 0, 2) and B (3, 9, 6) is given by which of the following?

- A.  $\underline{r} = \underline{i} + 2\underline{k} + \lambda(2\underline{i} + 9\underline{j} + 4\underline{k})$
- B.  $\underline{r} = \underline{i} + 2\underline{k} + \lambda(2\underline{i} + 9\underline{j} + 8\underline{k})$
- C.  $\underline{r} = \underline{i} + 2\underline{k} + \lambda(4\underline{i} + 9\underline{j} + 4\underline{k})$
- D.  $\underline{r} = \underline{i} + 2\underline{k} + \lambda(2\underline{i} + 4\underline{k})$

Q7. Which expression is equal to  $\int \tan^{-1} x \, dx$ ?

- A.  $x \tan^{-1} x - \int \frac{x}{1+x^2} dx$
- B.  $x^2 \tan^{-1} x - \int \frac{x^2}{1+x^2} dx$
- C.  $\frac{1}{2}(\tan^{-1} x)^2$
- D.  $\frac{x}{2}(\tan^{-1} x)^2 - \int \frac{x}{\tan x} dx$

Q8. Which of the following statements is FALSE?

- A.  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$  such that  $x^4 = 2y^3$
- B.  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$  such that  $x = y \Leftrightarrow \sin x = \sin y$
- C.  $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$  such that  $\sqrt{x} < e^y$
- D.  $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$  such that  $|y| > x$

Q9.  $\int_{-1}^2 2x\sqrt{x+2} \, dx$ .

- A.  $\frac{46}{15}$
- B.  $\frac{92}{15}$
- C.  $\frac{113}{2}$
- D.  $\frac{28+32\sqrt{2}}{15}$

Q10. If  $\int_a^b \frac{1}{\sin x + \cos x} \, dx = 1$ , where  $a, b \in \mathbb{R}$ ,

what is the value of  $\int_{2a}^{2b} \frac{1}{\sin \frac{x}{2} + \cos \frac{x}{2}} \, dx$

- A.  $\frac{1}{4}$
- B.  $\frac{1}{2}$
- C. 1
- D. 2

**End of Multiple Choice**

## Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

---

### Question 11

15 marks

a. Let  $z = 4cis\frac{5\pi}{6}$  and  $w = \sqrt{2}cis\frac{\pi}{4}$ .

(i) Find  $z^7$  in modulus-argument form.

2

(ii) Express  $\frac{z}{w}$  in both modulus-argument form and Cartesian form.

3

(iii) Hence or otherwise, find the exact value of  $\cos\frac{7\pi}{12}$ .

2

b. A point  $P$  moves on the  $x$ -axis so that its  $x$ -coordinate, in metres, at time  $t$  seconds is given by

$$x = 8[1 - \sin(2t)] + 3\cos(2t)$$

(i) Show that the motion of  $P$  is simple harmonic.

2

(ii) State centre and period of the motion of  $P$ .

2

c. Sketch the region on the Argand diagram defined by the following

$$\{z: 1 < |z| \leq 3\} \cap \{z: \operatorname{Im}(z) > \operatorname{Re}(z)\}$$

4

End of Question 11

- a. (i) Find the values of  $a, b$  and  $c$  such that 3

$$\frac{5x}{(x-1)(x^2+2x+2)} \equiv \frac{a}{x-1} + \frac{bx+c}{x^2+2x+2}$$

- (ii) Hence, or otherwise, find 3

$$\int \frac{5x}{(x-1)(x^2+2x+2)} dx.$$

- b. Let  $\omega = e^{i(\frac{\pi}{n})}$ , where  $\omega$  is a complex number ie.  $\omega$  is not  $\in \mathbb{R}$

- (i) Show that  $\omega$  is a solution of  $z^{2n} = 1$ . 1

- (ii) Given  $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$ ,  
deduce that  $1 + \omega + \dots + \omega^{2n-1} = 0$ . 1

- (iii) Show that  $1 + \omega + \dots + \omega^{n-1} = 1 + i \cot\left(\frac{\pi}{2n}\right)$ . 3

- c. Prove the following statement by contradiction. 4

*If  $a, b$  and  $c$  are odd integers and  $x \in \mathbb{Z}$ , then the equation  
 $ax^2 + bx + c = 0$  has no integer solution.*

**End of Question 12**

**Question 13****15 marks**

a. (i) Express  $1 + i \cot \theta$  in modulus-argument form, where  $0 < \theta < \frac{\pi}{2}$  **2**

(ii) Hence, or otherwise, simplify  $(1 + i \cot \theta)^{-3}$  **1**

b. Find the exact value of  $\int_0^{\frac{\pi}{2}} \frac{1}{2 - \sin x} dx$ . **4**

c. A sequence is defined recursively as  $u_1 = 1$ ,  $u_{n+1} = 3u_n + 2n + 1$ , for all positive integers  $n$ .

Use Mathematical Induction to prove that  $u_n = 3^n - n - 1$  for all positive integers  $n$ . **3**

d. A particle is initially at rest on the number line at a position of  $x = 1$ . The particle moves continuously along the number line according to the acceleration equation

$$\ddot{x} = \frac{4}{(x-2)^2} + \frac{8}{x^3}.$$

At time  $t$ , the velocity of the particle is  $v$ .

(i) Show that  $v^2 = -\frac{8}{x-2} - \frac{8}{x^2}$  **2**

(ii) Hence, or otherwise, determine the possible range of particle's displacement as it moves along the number line. **3**

**Question 13 completed.**

- a. Use an appropriate substitution to evaluate  $\int_0^4 \frac{\sqrt{x}}{(x\sqrt{x}+1)^2} dx$  3
- b. Consider the statement about two positive integers  $m$  and  $n$ .  
 “If  $m^2 + n^2$  is even, then either both  $m$  and  $n$  are even or both  $m$  and  $n$  are odd.”
- (i) Write down the contrapositive of the statement. 1
- (ii) Prove the original statement by contraposition. 2
- c. (i) If  $0 < a < 4$ , prove that  $\frac{1}{a} + \frac{1}{4-a} \geq 1$ . 2
- (ii) If  $0 < a < 4, 0 < b < 4$  and  $0 < c < 4$ , prove that at least one of the numbers
- $$\frac{1}{a} + \frac{1}{4-b}, \frac{1}{b} + \frac{1}{4-c}, \frac{1}{c} + \frac{1}{4-a}$$
- is greater than or equal to one. 3
- d. Let  $I_n = \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx$ , where  $n = 1, 2, 3, \dots$  4
- Deduce that  $2nI_n + (2n-1)I_{n-1} = \sqrt{2}$ , where  $n = 2, 3, 4, \dots$

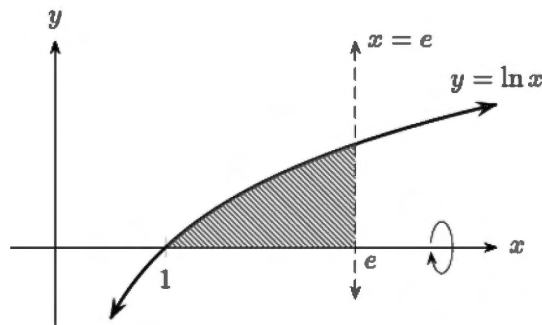
**Question 14 completed.**

**Question 15**

**15 marks**

- a. The region enclosed by the curve  $y = \ln x$ , the line  $x = e$  and the  $x$  axis is shown in the diagram.

**3**



Find the volume of the solid formed by rotating the shaded region about the  $x$  axis.

- b. Sphere  $S$  has a vector equation

$$\left| \vec{r} - (3\vec{i} + \vec{j} + 4\vec{k}) \right| = \sqrt{35}$$

- (i) Write the Cartesian equation of this sphere.

**1**

- (ii) The line  $l$  has the equation  $\vec{r} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ ,  $\lambda \in \mathbb{R}$ .

**3**

Determine whether this line is a tangent to the sphere  $S$  or not. Give full reasons for your answer.

- c. Show there exists complex numbers  $z$  and  $w$ , such that

**3**

$$\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w} \text{ where } z \neq 0 \text{ and } w \neq 0.$$

- d. (i) Show  $a^2 + b^2 + c^2 \geq ab + ac + bc$

**1**

- (ii) Hence, show  $(a+b+c)^2 \geq 3(ab+ac+bc)$

**1**

- (iii) Hence, prove  $a^2 + b^2 + c^2 \geq 3 \times \sqrt[3]{a^2 b^2 c^2}$

**3**

$$(\text{Given } a^3 + b^3 = (a+b)(a^2 - ab + b^2))$$

**Question 15 completed**

**Question 16**

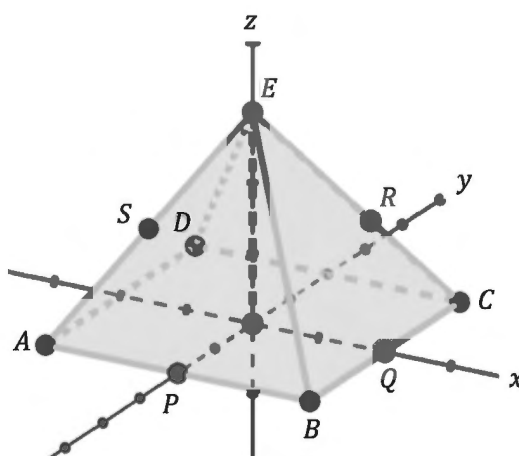
**15 marks**

- a. (i) Solve  $z^3 = -1$ , giving your answers in modulus-argument form. **2**

- (ii) Using part (i), or otherwise, find the solutions to the equation **3**

$$(z^2 + 2)^3 = -1$$

- b. ABCDE is a pyramid with base measuring 8 cm by 8 cm.  
The height of the pyramid is  $4e$ , where  $e$  is a real positive number.  
The origin  $O$  is the centre of the base.



The points  $P, Q, R$  and  $S$  are the midpoints of sides  $AB, BC, CE$  and  $EA$  respectively.

The position vectors of  $A, B, C$  and  $D$  relative to  $O$ , respectively, are

$$-4\underset{\sim}{i} - 4\underset{\sim}{j}, 4\underset{\sim}{i} - 4\underset{\sim}{j}, 4\underset{\sim}{i} + 4\underset{\sim}{j} \text{ and } -4\underset{\sim}{i} + 4\underset{\sim}{j}.$$

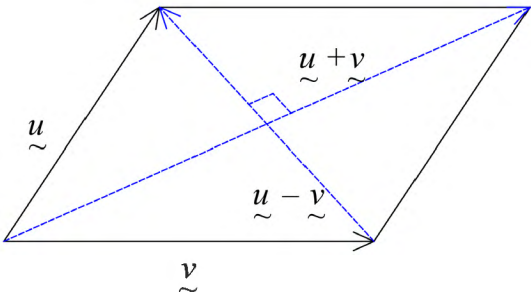
- (i) Show, in terms of  $e$ , that  $\overrightarrow{PR} = 2\underset{\sim}{i} + 6\underset{\sim}{j} + 2e\underset{\sim}{k}$  and  $\overrightarrow{QS} = -6\underset{\sim}{i} - 2\underset{\sim}{j} + 2e\underset{\sim}{k}$ . **3**

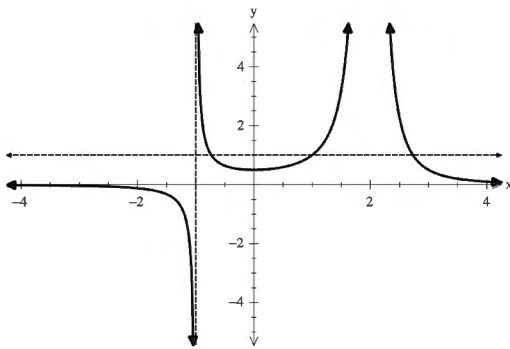
- (ii) Hence or otherwise, write the vector equations of line  $l$ , passing through points  $P$  and  $R$ , and line  $s$ , passing through points  $Q$  and  $S$ . **2**

- (iii) Find the position vector  $\overrightarrow{OM}$  in terms of  $e$ , where  $M$  is the point of intersection of lines  $l$  and  $s$ . **2**

- (iv) Given that  $\overrightarrow{OM}$  is perpendicular to  $\overrightarrow{EB}$ , find the value of  $e$  and hence find the acute angle between  $\overrightarrow{PR}$  and  $\overrightarrow{QS}$ . Give your answer to the nearest degree. **3**

**End of paper**

|    |                                                                                                                                                                                                                                                                                                                                                                       |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q1 | $z^3 - 4z^2 + 6z - 4 = 0$ <p>roots are <math>(1 - i)</math> and <math>(1 + i)</math></p> <p>factor <math>z^2 - 2\operatorname{Re}(\alpha)z +  \alpha ^2</math></p> $\therefore z^2 - 2z + 2$ $z^3 - 4z^2 + 6z - 4 = (z^2 - 2z + 2)(az + b)$ $\therefore \begin{aligned} a &= 1 \\ b &= -2 \end{aligned}$ <p>Other roots are <math>1 + i</math> and <math>2</math></p> | B |
| Q2 | $u = 2i - aj - k$ $v = 3i + 2j - bk$ $u \cdot v = 0$ $u \cdot v = 2 \times 3 - 2a + b = 0$ $b - 2a = -6$ $-8 - 2(-1) = -6$ $a = -1 \quad b = -8$                                                                                                                                                                                                                      | D |
| Q3 |  <p>Diagonals of rhombus are perpendicular therefore</p> $(\underline{\underline{u}} + \underline{\underline{v}}) \cdot (\underline{\underline{u}} - \underline{\underline{v}}) = 0$                                                                                               | C |

|    |                                                                                                                                                                                                                                                                                                                                                                       |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q4 |                                                                                                                                                                                                                                                                                      | D |
| Q5 | If the soup contains chilli, the soup does not contain lemon nor vinegar.                                                                                                                                                                                                                                                                                             | A |
| Q6 | $A = \underset{\sim}{i} + 2\underset{\sim}{k}$ $B = 3\underset{\sim}{i} + 9\underset{\sim}{j} + 6\underset{\sim}{k}$ $\overrightarrow{AB} = 2\underset{\sim}{i} + 9\underset{\sim}{j} + 4\underset{\sim}{k}$ $\underset{\sim}{r} = \underset{\sim}{i} + 2\underset{\sim}{k} + \lambda \left( 2\underset{\sim}{i} + 9\underset{\sim}{j} + 4\underset{\sim}{k} \right)$ | A |
| Q7 | $u = \tan^{-1} x \qquad v' = 1$ $u' = \frac{1}{1+x^2} \qquad v = x$ $I = uv - \int vu' dx$ $= x \tan^{-1} x - \int \frac{x}{1+x^2} dx$                                                                                                                                                                                                                                | A |
| Q8 | $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x = y \Leftrightarrow \sin x = \sin y$                                                                                                                                                                                                                                                                | B |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q9  | $\int_{-1}^2 2x\sqrt{x+2} \, dx$ <p> <math>u = x + 2</math><br/> <math>x = u - 2 \Rightarrow dx = du</math><br/> <math>x = -1 \Rightarrow u = 1</math><br/> <math>x = 2 \Rightarrow u = 4</math> </p> $\int_1^4 2(u-2)\sqrt{u} \, du$ $= 2 \int_1^4 u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \, du$ $= 2 \left[ \frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} \right]_1^4$ $= 2 \left\{ \left( \frac{2}{5} \times 2^5 - \frac{4}{3} \times 2^3 \right) - \left( \frac{2}{5} \times 1 - \frac{4}{3} \times 1 \right) \right\}$ $= 2 \left( \frac{64}{5} - \frac{32}{3} - \frac{2}{5} + \frac{4}{3} \right) = \frac{92}{15}$ | A |
| Q10 | $\int_a^b \frac{1}{\sin x + \cos x} \, dx = 1$ <p> <math>\text{let } x = \frac{u}{2} \Rightarrow 2x = u</math><br/> <math>2dx = du</math><br/> <math>x = a \Rightarrow u = 2a</math><br/> <math>x = b \Rightarrow u = 2b</math> </p> $\therefore \int_{2a}^{2b} \frac{2}{\sin \frac{u}{2} + \cos \frac{u}{2}} \, du = 1$ $\int_{2a}^{2b} \frac{1}{\sin \frac{x}{2} + \cos \frac{x}{2}} \, dx = \frac{1}{2}$                                                                                                                                                                                                                      | B |

Question 11.

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a-i   | $z = R\text{cis}\theta$ $z^7 = R^7\text{cis}(7\theta)$ $\therefore z^7 = 4^7\text{cis}\frac{35\pi}{6}$ $= 16384\text{cis}\left(-\frac{\pi}{6}\right)$                                                                                                                                                                                                                                                                                                                                                                                  | <p>2 marks – correct solution</p> <p>1 mark – either</p> <ul style="list-style-type: none"> <li>- correct argument</li> <li>- correct modulus</li> </ul>                                                                                                                                                                                                                                                  |
| a-ii  | <p>ModArg form</p> $\frac{z_1}{z_2} = \frac{R_1\text{cis}\theta}{R_2\text{cis}\beta} = \frac{R_1}{R_2}\text{cis}(\theta - \beta)$ $= \frac{4}{\sqrt{2}}\text{cis}\left(\frac{5\pi}{6} - \frac{\pi}{4}\right)$ $= 2\sqrt{2}\text{cis}\frac{7\pi}{12}$ <p>Cartesian form:</p> $\frac{z}{w} = \frac{-2\sqrt{3} + 2i}{1 + i} \times \frac{1 - i}{1 - i} = (1 - \sqrt{3}) + i(1 + \sqrt{3})$ $\frac{z}{w} = \frac{-2\sqrt{3} + 2}{2} + \frac{2 + 2\sqrt{3}}{2}i \quad \text{was also accepted}$ <p>Decimal Form was <i>NOT</i> accepted</p> | <p>3 marks Provides correct solution</p> <p>2 marks</p> <p>Expresses fraction in both modulus-argument or Cartesian form (but with the denominator not realised), or equivalent merit</p> <p>1 mark</p> <ul style="list-style-type: none"> <li>- uses <math>\frac{z_1}{z_2}</math> to produce correct modulus or argument</li> <li>- uses <math>\frac{z}{w}</math> to produce correct fraction</li> </ul> |
| a-iii | <p>Using part (ii) and equating the real parts,</p> $2\sqrt{2}\cos\frac{7\pi}{12} = 1 - \sqrt{3}$ $\therefore \cos\frac{7\pi}{12} = \frac{1 - \sqrt{3}}{2\sqrt{2}}$                                                                                                                                                                                                                                                                                                                                                                    | <p>2 marks – correct solution</p> <p>1 mark - Attempts to equate real or imaginary parts of the two expression in part (i), or equivalent merit</p>                                                                                                                                                                                                                                                       |

|      |                                                                                                                                                                                                                                                                                                                                                            |                                                                                               |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| c-i  | <p>Note:</p> $x = 8[1 - \sin(2t)] + 3\cos(2t)$ $\therefore x - 8 = -8\sin(2t) + 3\cos(2t)$ $\dot{x} = -16\cos(2t) - 6\sin(2t)$ $\ddot{x} = 32\sin(2t) - 12\cos(2t)$ $= -4(x - 8) \text{ which is in the form } -n^2(x - c)$ <p>Hence motion is SHM</p>                                                                                                     | <p>2 marks – correct solution</p> <p>1 mark – correct approach with one arithmetic error.</p> |
| c-ii | <p>Centre <math>x = 8</math>, period <math>= \frac{2\pi}{2} = \pi</math></p>                                                                                                                                                                                                                                                                               | <p>2 marks – both answers</p> <p>1 mark – one answer only</p>                                 |
| d    | <div data-bbox="400 622 791 994" data-label="Figure"> </div> <p>Features to include</p> <ol style="list-style-type: none"> <li>1. Correct circles and shading.</li> <li>2. Correct boundaries ie. full or dotted line</li> <li>3. Correct angle or indicated line is <math>y = x</math> or equivalent,</li> <li>4. Open circles used correctly.</li> </ol> | <p>4 marks – one mark for each correct feature</p>                                            |

Question 12

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a-i  | $5x \equiv a(x^2 + 2x + 2) + (x - 1)(bx + c)$ <p>Let</p> $\begin{array}{lll} x = 1: & 5 = 5a & \Rightarrow a = 1 \\ x = 0: & 0 = 2a - c & \Rightarrow c = 2 \\ x = -1: & -5 = a - 2(-b + c) & \Rightarrow b = -1 \end{array}$                                                                                                                                                                                                                                 | <p>3marks – correct solution</p> <p>2 marks – correct approach with one arithmetic error.</p> <p>1 mark – correct initial equation equating numerators</p>                                                                                                                                                                                                                                               |
| a-ii | $I = \int \frac{5x}{(x-1)(x^2+2x+2)} dx = \int \left( \frac{1}{x-1} + \frac{-x+2}{x^2+2x+2} \right) dx$ $I = \int \left( \frac{1}{x-1} - \frac{\frac{1}{2}(2x+2)-3}{x^2+2x+2} \right) dx$ $I = \int \left( \frac{1}{x-1} - \frac{1}{2} \times \frac{2x+2}{x^2+2x+2} + \frac{3}{(x+1)^2+1} \right) dx$ $I = \ln x-1  - \frac{1}{2} \ln(x^2+2x+2) + 3 \tan^{-1}(x+1) + c$ <p>or</p> $I = \ln \left  \frac{x-1}{\sqrt{x^2+2x+2}} \right  + 3 \tan^{-1}(x+1) + c$ | <p>3 marks – correct solution</p> <p>2 marks –</p> <ul style="list-style-type: none"> <li>- writes the integrand as the correct sum of fractions and completes the square in a denominator, or equivalent merit</li> </ul> <p>1 mark</p> <ul style="list-style-type: none"> <li>- correct integration achieving <math>\ln x-1 </math></li> <li>- attempts to write as sum of three fractions.</li> </ul> |
| b-i  | $\begin{aligned} \omega^{2n} &= \left( e^{\pi i/n} \right)^{2n} \\ &= e^{2\pi i} \\ &= 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                       | <p>1 mark - Provides correct solution</p>                                                                                                                                                                                                                                                                                                                                                                |
| b-ii | $\begin{aligned} \omega^{2n} - 1 &= 0 \\ \Rightarrow (\omega - 1)(\omega^{2n-1} + \dots + \omega + 1) &= 0 \\ \text{Since } \omega \neq 1, \omega^{2n-1} + \dots + \omega + 1 &= 0 \end{aligned}$                                                                                                                                                                                                                                                             | <p>1 mark - Provides correct solution including statement that <math>\omega \neq 1</math></p>                                                                                                                                                                                                                                                                                                            |

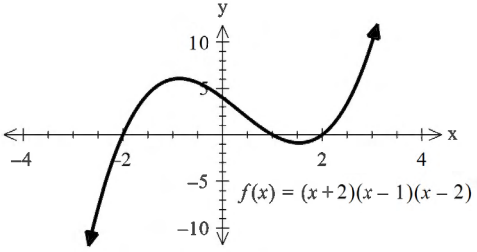
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                   |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| b-iii | $1 + \omega + \dots + \omega^{n-1} = \frac{1 - \omega^n}{1 - \omega}$ $= \frac{1 - e^{\pi i}}{1 - \cos\left(\frac{\pi}{n}\right) - i \sin\left(\frac{\pi}{n}\right)}$ <div data-bbox="387 286 531 360" style="border: 1px solid black; padding: 2px; display: inline-block;">Line 1</div> $= \frac{1 - (-1)}{1 - \cos\left(\frac{\pi}{n}\right) - i \sin\left(\frac{\pi}{n}\right)} \times \frac{1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)}{1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)}$ $= \frac{2 \left(1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)\right)}{\left(1 - \cos\left(\frac{\pi}{n}\right)\right)^2 + \left(\sin\left(\frac{\pi}{n}\right)\right)^2}$ <div data-bbox="383 571 525 645" style="border: 1px solid black; padding: 2px; display: inline-block;">Line 2</div> $= \frac{2 \left(1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)\right)}{2 - 2 \cos\left(\frac{\pi}{n}\right)}$ $= \frac{1 - \cos\left(\frac{\pi}{n}\right)}{1 - \cos\left(\frac{\pi}{n}\right)} + i \frac{\sin\left(\frac{\pi}{n}\right)}{1 - \cos\left(\frac{\pi}{n}\right)}$ $= 1 + i \frac{2 \sin\left(\frac{\pi}{2n}\right) \cos\left(\frac{\pi}{2n}\right)}{2 \sin^2\left(\frac{\pi}{2n}\right)}$ $= 1 + i \cot\left(\frac{\pi}{2n}\right)$ | <p>3 marks - Provides correct solution</p> <p>2 marks - correct to Line 2</p> <p>1 mark – Produces either fraction in Line 1.</p> |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| c | <p>Suppose for a contradiction that <math>a</math>, <math>b</math> and <math>c</math> are odd integers and the equation <math>ax^2 + bx + c = 0</math> has an integer solution.</p> <p>For <math>A, B, C \in \mathbb{Z}</math>, let<br/> <math>a = 2A + 1</math>,<br/> <math>b = 2B + 1</math>, and<br/> <math>c = 2C + 1</math><br/> <math>\therefore (2A + 1)x^2 + (2B + 1)x + (2C + 1) = 0</math> [*]</p> <p><b>Case 1:</b> <math>x</math> is an even number, i.e. <math>x = 2k</math>, <math>k \in \mathbb{Z}</math>.<br/> Since <math>x</math> is a solution, substitute into [*]:<br/> <math>(2A + 1)(2k)^2 + (2B + 1)(2k) + (2C + 1) = 0</math><br/> Expand to give:<br/> <math>2[2k^2(2A + 1) + k(2B + 1) + C] + 1 = 0</math><br/> LHS is an odd number which cannot equal to zero, hence absurd.</p> <p><b>Case 2:</b> <math>x</math> is an odd number, i.e. <math>x = 2k + 1</math>, <math>k \in \mathbb{Z}</math>.<br/> Since <math>x</math> is a solution, substitute into [*]:<br/> <math>(2A + 1)(2k + 1)^2 + (2B + 1)(2k + 1) + (2C + 1) = 0</math><br/> Expand to give:<br/> <math>(2A + 1)(4k^2 + 4k + 1) + 4Bk + 2B + 2k + 1 + 2C + 1 = 0</math><br/> <math>2A(4k^2 + 4k) + 4k^2 + 6k + 3 + 4Bk + 2B + 2C = 0</math><br/> <math>2[A(4k^2 + 4k) + 2k^2 + 3k + 1 + 2Bk + B + C] + 1 = 0</math><br/> LHS is an odd number which cannot equal to zero, hence absurd.</p> <p><math>\therefore</math> initial supposition is false<br/> <math>\therefore</math> if <math>a</math>, <math>b</math> and <math>c</math> are odd integers, then<br/> <math>ax^2 + bx + c = 0</math> has no integer solution</p> | <p>4 marks – correct solution.</p> <p>3 marks – attempts both odd and even expressions but only one correct.</p> <p>2 marks</p> <ul style="list-style-type: none"> <li>- attempts one case only and proven correctly.</li> <li>- attempts to use quadratic formula to show no integer solutions ie. more than simply deriving quad formula</li> </ul> <p>1 mark – correctly converts <math>a</math>, <math>b</math> and <math>c</math> to an expression for odd numbers.</p> |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Question 13

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a-i  | $  \begin{aligned}  1 + i \cot \theta &= 1 + i \frac{\cos \theta}{\sin \theta} \\  &= \frac{\sin \theta + i \cos \theta}{\sin \theta} \\  &= \frac{1}{\sin \theta} [\sin \theta + i \cos \theta] \\  &= \frac{1}{\sin \theta} \left[ \cos \left( \frac{\pi}{2} - \theta \right) + i \sin \left( \frac{\pi}{2} - \theta \right) \right] \\  &= \operatorname{cosec} \theta \operatorname{cis} \left( \frac{\pi}{2} - \theta \right)  \end{aligned}  $ <p><math>\sqrt{1 + \cot^2 \theta}</math> was also accepted. If the argument was in degrees this was also accepted.</p>                                                                                                     | <p>2 marks – correct solution</p> <p>1 mark – either</p> <ul style="list-style-type: none"> <li>- correct modulus</li> <li>- correct argument</li> </ul>                                                                                                                                                        |
| a-ii | $  \begin{aligned}  (1 + i \cot \theta)^{-3} &= \left( \operatorname{cosec} \theta \operatorname{cis} \left( \frac{\pi}{2} - \theta \right) \right)^{-3} \\  &= \operatorname{cosec}^{-3} \theta \operatorname{cis} \left[ -3 \left( \frac{\pi}{2} - \theta \right) \right] \\  &= \sin^3 \theta \operatorname{cis} \left( 3\theta - \frac{3\pi}{2} \right) \\  &= \sin^3 \theta \operatorname{cis} \left( 3\theta + \frac{\pi}{2} \right)  \end{aligned}  $                                                                                                                                                                                                                    | 1 mark                                                                                                                                                                                                                                                                                                          |
| b    | <p>Let <math>t = \tan \frac{x}{2}</math>    <math>dt = \frac{1}{2} \sec^2 \frac{x}{2} dx = \frac{1}{2} (1 + t^2) dx</math>    <math>dx = \frac{2}{1+t^2} dt</math></p> <p><math>\therefore \sin x = \frac{2t}{1+t^2}</math><br/> When <math>x = 0, t = 0</math><br/> <math>x = \frac{\pi}{2}, t = 1</math></p> <p><math>\therefore I = \int_0^{\frac{\pi}{2}} \frac{1}{2 - \sin x} dx = \int_0^1 \frac{1}{2 - \frac{2t}{1+t^2}} \frac{2}{1+t^2} dt</math></p> <p><math>I = \int_0^1 \frac{2}{2 - \frac{2t}{1+t^2}} \times \frac{2}{1+t^2} dt</math></p> <p><math>I = \int_0^1 \frac{2}{2(1+t^2) - 2t} dt</math></p> <p><math>I = \int_0^1 \frac{2}{2t^2 - 2t + 2} dt</math></p> | <p>4 marks - Provides correct solution</p> <p>3 marks - Finds the correct integral, or equivalent merit</p> <p>2 marks - Completes <math>t</math>-substitution, including the limits of integration, or equivalent merit</p> <p>1 mark - Attempts to use a <math>t</math>-substitution, or equivalent merit</p> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | $= \int_0^1 \frac{1}{t^2 - t + 1} dt$ $= \int_0^1 \frac{1}{\left(t - \frac{1}{2}\right)^2 + \frac{3}{4}} dt$ $= \left[ \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2t - 1}{\sqrt{3}} \right) \right]_0^1$ $= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} - \frac{2}{\sqrt{3}} \tan^{-1} \left( -\frac{1}{\sqrt{3}} \right)$ $= \frac{2}{\sqrt{3}} \times \frac{\pi}{6} - \frac{2}{\sqrt{3}} \times -\frac{\pi}{6}$ $= \frac{4\pi}{6\sqrt{3}}$ $= \frac{2\pi}{3\sqrt{3}}$ $= \frac{2\sqrt{3}\pi}{9}$ |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| c    | <p><math>P(n): u_n = 3^n - n - 1, n \in \mathbb{Z}^+,</math><br/>for the sequence <math>u_1 = 1, u_{n+1} = 3u_n + 2n + 1, n \in \mathbb{Z}^+.</math></p> <p>Prove <math>P(1)</math> true:<br/>By formula, <math>u_1 = 3^1 - 1 - 1 = 1</math><br/>By definition, <math>u_1 = 1</math><br/><math>\therefore P(1)</math> is true</p> <p>Assume <math>P(k)</math> true, <math>k \in \mathbb{Z}^+</math><br/>i.e. <math>u_k = 3^k - k - 1</math></p> <p>Prove <math>P(k+1)</math> true.<br/>i.e. prove <math>u_{k+1} = 3^{k+1} - (k+1) - 1 = 3^{k+1} - k - 2</math><br/>Now, from the definition:<br/><math>u_{k+1} = 3u_k + 2k + 1</math><br/><math>u_{k+1} = 3(3^k - k - 1) + 2k + 1</math> using the assumption<br/><math>u_{k+1} = 3^{k+1} - 3k - 3 + 2k + 1</math><br/><math>u_{k+1} = 3^{k+1} - k - 2</math><br/><math>\therefore P(k+1)</math> is true if <math>P(k)</math> is true.</p> <p>Hence, by mathematical induction, <math>P(n)</math> is true for all positive integers <math>n \geq 1</math></p> | <p>3 marks – correct solution</p> <p>2 marks - Proves true for <math>n = 1</math> and incorporates the assumption <math>P(k)</math> into <math>P(k+1)</math></p> <p>1 mark - Proves true for <math>n = 1</math></p> |
| d-i  | $\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 4(x-2)^{-2} + 8x^{-3}$ $\frac{1}{2} v^2 = -4(x-2)^{-1} - 4x^{-2} + c$ $x=1, v=0 \Rightarrow c=0$ $\therefore \frac{1}{2} v^2 = -4(x-2)^{-1} - 4x^{-2}$ $\therefore v^2 = -\frac{8}{x-2} - \frac{8}{x^2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>2 marks - Provides correct solution</p> <p>1 mark - Makes some use of <math>\frac{d}{dx} \left( \frac{1}{2} v^2 \right)</math> or equivalent merit</p>                                                           |
| d-ii |  <p><math>v^2 \geq 0</math></p> $\Rightarrow -8 \left( \frac{1}{x-2} + \frac{1}{x^2} \right) \geq 0$ $\Rightarrow \frac{1}{x-2} + \frac{1}{x^2} \leq 0$ $\Rightarrow \frac{x^2 + x - 2}{x^2(x-2)} \leq 0$ $\Rightarrow \frac{(x+2)(x-1)(x-2)}{x^2(x-2)^2} \leq 0$ <p>From the sketch of the cubic numerator,</p> <p><math>x \leq -2</math> or <math>1 \leq x &lt; 2</math> [Note that <math>x-2</math> is originally in the denominator]</p> <p>Since the particle moves continuously, we can only use <math>1 \leq x &lt; 2</math></p>                                                                                                                                                                                                                                                                                                                                                                                    | <p>3 marks – correct solution</p> <p>2 marks – determine both possible solutions but does not exclude <math>x=2</math></p> <p>1 mark – attempts to solve required inequality.</p>                                   |

OR

$$\frac{x^2 + x - 2}{x^2(x - 2)} = 0 \quad x \neq 0 \text{ or } x \neq 2$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2 \text{ or } x = 1$$

Test points: when  $x < -2$  inequality is true

when  $-2 < x < 0$  inequality is false

when  $0 < x < 1$  inequality is false

when  $1 < x < 2$  inequality is true

when  $x > 2$  inequality is false

since the particle starts at  $x = 1$  and moves continuously  
only solution is  $1 \leq x < 2$

However, students who left  $x \leq -2$  were not penalised.

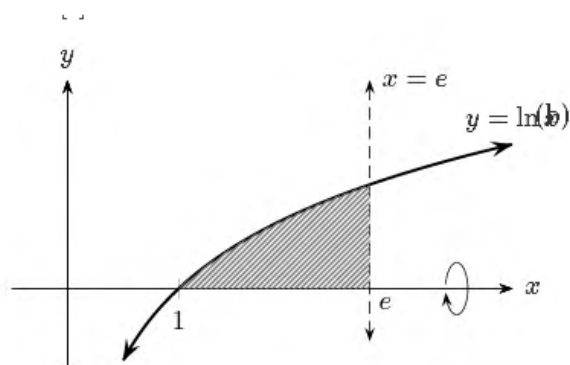
Question 14

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a    | $I = \int_0^4 \frac{\sqrt{x}}{(x\sqrt{x} + 1)^2} dx$ <p>Let <math>u^2 = x</math> (<math>u \geq 0</math>) è <math>2u du = dx</math><br/> When <math>x = 0, u = 0</math><br/> <math>x = 4, u = 2</math></p> $\therefore I = \int_0^2 \frac{u}{(u^2 \cdot u + 1)^2} 2u du = \int_0^2 \frac{2u^2}{(u^3 + 1)^2} du$ $I = \left[ -\frac{2}{3(u^3 + 1)} \right]_0^2 = -\frac{2}{3 \times 9} - \left( -\frac{2}{3} \right) = \frac{16}{27}$                                                                                                                                                                               | <p>3 marks - correct solution</p> <p>2 marks - Obtains a correct primitive, or equivalent merit</p> <p>1 mark - Uses an appropriate substitution, including the limits of integration, or equivalent merit</p> |
| b-i  | <p>If <math>x, y</math> are two integers for which one is odd and one is even, then <math>x + y</math> is odd.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>1 mark – correct solution</p>                                                                                                                                                                               |
| b-ii | <p>Assume <math>x</math> is even, <math>y</math> is odd</p> $\exists k, m \in \mathbb{Z} \text{ such } x = 2k, y = 2m + 1$ $x + y = 2k + 2m + 1$ $= 2(k + m) + 1$ <p><math>\therefore x + y</math> is odd</p> <p><math>nQ \Rightarrow nP</math></p>                                                                                                                                                                                                                                                                                                                                                               | <p>2 marks – correct solution</p> <p>1 mark – attempts correct method ie. makes correct statement</p> <p><math>\exists k, m \in \mathbb{Z} \text{ such } x = 2k, y = 2m + 1</math></p>                         |
| c-i  | $\frac{1}{a} + \frac{1}{4-a} - 1 = \frac{4-a+a-a(4-a)}{a(4-a)}$ $= \frac{(a-2)^2}{a(4-a)} \geq 0 \text{ since } 0 < a < 4 \text{ makes the denominator positive}$ <p>Hence <math>\frac{1}{a} + \frac{1}{4-a} \geq 1</math></p>                                                                                                                                                                                                                                                                                                                                                                                    | <p>2 marks – correct solution</p> <p>1 mark – forms correct inequality</p>                                                                                                                                     |
| c-ii | <p>Assume <math>\frac{1}{a} + \frac{1}{4-b} &lt; 1, \frac{1}{b} + \frac{1}{4-c} &lt; 1</math> and <math>\frac{1}{c} + \frac{1}{4-a} &lt; 1</math></p> <p>By addition,</p> $\left( \frac{1}{a} + \frac{1}{4-b} \right) + \left( \frac{1}{b} + \frac{1}{4-c} \right) + \left( \frac{1}{c} + \frac{1}{4-a} \right) < 3$ $\Rightarrow \left( \frac{1}{a} + \frac{1}{4-a} \right) + \left( \frac{1}{b} + \frac{1}{4-b} \right) + \left( \frac{1}{c} + \frac{1}{4-c} \right) < 3$ <p>But, from part (i), each bracket is at least 1, a contradiction.</p> <p>Hence, at least one of the three numbers is at least 1</p> | <p>3 marks – correct solution</p> <p>2 marks - Makes significant progress towards a solution.</p> <p>1 mark - Makes little progress towards a solution</p>                                                     |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                     |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Option 1</p> $I_n = \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx$ $I_n = \int_0^1 \frac{x^2 x^{2n-2}}{\sqrt{1+x^2}} dx \text{ since } n \geq 2$ $I_n = \int_0^1 \frac{(1+x^2-1)x^{2n-2}}{\sqrt{1+x^2}} dx$ $I_n = \int_0^1 \frac{(1+x^2)x^{2n-2}}{\sqrt{1+x^2}} dx - \int_0^1 \frac{x^{2(n-1)}}{\sqrt{1+x^2}} dx$ $I_n = \int_0^1 x^{2n-2} \sqrt{1+x^2} dx - I_{n-1}$ <p>d For <math>\int_0^1 x^{2n-2} \sqrt{1+x^2} dx</math>, integrate by parts as follows:</p> $u = \sqrt{1+x^2} \quad \therefore \frac{du}{dx} = \frac{x}{\sqrt{1+x^2}}$ $\frac{dv}{dx} = x^{2n-2} \quad \therefore v = \frac{1}{2n-1} x^{2n-1}$ $\therefore I_n = \left[ \frac{x^{2n-1} \sqrt{1+x^2}}{2n-1} \right]_0^1 - \frac{1}{2n-1} \int_0^1 x^{2n-1} \frac{x}{\sqrt{1+x^2}} dx - I_{n-1}$ $I_n = \frac{\sqrt{2}}{2n-1} - \frac{1}{2n-1} \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx - I_{n-1}$ $I_n = \frac{\sqrt{2}}{2n-1} - \frac{1}{2n-1} I_n - I_{n-1}$ $(2n-1)I_n = \sqrt{2} - I_n - (2n-1)I_{n-1}$ $\therefore 2n I_n + (2n-1)I_{n-1} = \sqrt{2} \text{ as required}$ | <p>4 marks – correct solution.</p> <p>3 marks - Makes substantial progress, identifying the terms <math>I_n</math> and <math>I_{n-1}</math></p> <p>2 marks - Attempts to use integration by parts and attempts a manipulation to obtain <math>I_{n-1}</math> or <math>I_n</math>, or equivalent merit</p> <p>1 mark - Attempts to use integration by parts, or equivalent merit</p> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| d. | <p>Option 2</p> $I_n = \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx$ $= \int_0^1 \frac{x \cdot x^{2n-1}}{\sqrt{1+x^2}} dx$ $u = x^{2n-1} \quad dv = \frac{x}{\sqrt{1+x^2}}$ $du = (2n-1)x^{2n-2} \quad v = \int \frac{x}{\sqrt{1+x^2}} dx$ $= \frac{1}{2} \int \frac{2x}{\sqrt{1+x^2}} dx$ $= (1+x^2)^{\frac{1}{2}} = \sqrt{1+x^2}$ $I_n = \left[ x^{2n-1} \sqrt{1+x^2} \right]_0^1 - \int_0^1 \sqrt{1+x^2} \cdot (2n-1) \cdot x^{2n-2} dx$ $= \sqrt{2} - \int_0^1 \sqrt{1+x^2} \cdot (2n-1) \cdot x^{2n-2} dx$ $= \sqrt{2} - (2n-1) \cdot \int_0^1 \sqrt{1+x^2} \cdot x^{2n-2} dx$ $= \sqrt{2} - (2n-1) \cdot \int_0^1 \frac{(1+x^2) \cdot x^{2n-2}}{\sqrt{1+x^2}} dx$ $= \sqrt{2} - (2n-1) \left[ \int_0^1 \frac{x^{2n-2}}{\sqrt{1+x^2}} dx + \int_0^1 \frac{x^2}{\sqrt{1+x^2}} dx \right]$ $= \sqrt{2} - (2n-1)[I_{n-1} + I_n]$ $= \sqrt{2} - (2n-1)I_{n-1} - (2n-1)I_n]$ $I_n + 2nI_n - I_n + (2n-1)I_{n-1} = \sqrt{2}$ $2nI_n + (2n-1)I_{n-1} = \sqrt{2}$ |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

Question 15



$$V = \pi \int_1^e y^2 dx$$

$$= \pi \int_1^e (\ln x)^2 dx$$

Finding the integral by parts/inserting  
'phantom' term

a

$$\left| \begin{array}{ll} u = (\ln x)^2 & dv = 1 \\ du = 2 \ln x \times \frac{1}{x} & v = x \\ & = \frac{2}{x} \ln x \end{array} \right.$$

$$V = \pi \left( \left[ x (\ln x)^2 \right]_1^e - \int_1^e \frac{2}{x} \ln x \times x dx \right)$$

$$= \pi \left( e - 2 \int_1^e \ln x dx \right)$$

Use the area about the  $y$  axis:

$$\left| \begin{array}{l} \int_1^e \ln x dx = (e \times 1) - \int_{y=0}^{y=1} e^y dy \\ = e - [e - 1] \\ = 1 \\ \therefore V = \pi(e - 2) \text{ units}^3 \end{array} \right.$$

3 marks – correct solution

2 marks – correct use of  
IBP

1 mark

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| b-i | <p>Let <math>\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}</math>,</p> $\left  \underline{r} - (3\underline{i} + \underline{j} + 4\underline{k}) \right  = \sqrt{35}$ $\left  (x-3)\underline{i} + (y-1)\underline{j} + z-4\underline{k} \right  = \sqrt{35}$ $\therefore (x-3)^2 + (y-1)^2 + (z-4)^2 = 35$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 mark  |
| b-i | $\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ <p>If the line is a tangent, then only one unique value of <math>\lambda</math> exists for the expression</p> $\left  \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right  = \sqrt{35}$ <p>Test for values of <math>\lambda</math> that satisfy following</p> $\left  \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right  = \sqrt{35}$ $\text{LHS} = \left  \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right $ $= \left  \begin{pmatrix} -1 + \lambda \\ -1 + 2\lambda \\ -1 - \lambda \end{pmatrix} \right $ $= \sqrt{(\lambda - 1)^2 + (2\lambda - 1)^2 + (-1 - \lambda)^2}$ $= \sqrt{\lambda^2 - 2\lambda + 1 + 4\lambda^2 - 4\lambda + 1 + \lambda^2 + 2\lambda + 1}$ $= \sqrt{6\lambda^2 - 4\lambda + 3} = \sqrt{35}$ $6\lambda^2 - 4\lambda + 3 = 35$ $6\lambda^2 - 4\lambda - 32 = 0$ $3\lambda^2 - 2\lambda - 16 = 0$ $\Delta = (-2)^2 - 4(-16)(3)$ $= 196 > 0$ <p><math>\therefore</math> 2 solutions <math>\Rightarrow</math> not a tangent</p> | 3 marks |

Show  $z, w$  exist given  $\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w}$

$\therefore$

$$\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w}$$

$$\frac{z+w}{z} + \frac{z+w}{w} = 1$$

$$\frac{z}{z} + \frac{w}{z} + \frac{z}{w} + \frac{w}{w} = 1$$

$$1 + \frac{z}{w} + \frac{w}{z} + 1 = 1$$

$$\frac{z}{w} + \frac{w}{z} + 1 = 0 \quad \therefore \text{Line 1}$$

Let  $k = \left(\frac{z}{w}\right)$

Line 1 becomes

$$k + \frac{1}{k} + 1 = 0$$

$$k^2 + 1 + k = 0$$

$$k^2 + k + 1 = 0$$

$$k^2 + k = -1$$

$$k^2 + k + \frac{1}{4} + \frac{3}{4} = 0$$

$$\left(k + \frac{1}{2}\right)^2 + \frac{3}{4} = 0$$

$$\left(k + \frac{1}{2}\right)^2 = -\frac{3}{4}$$

$$k = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$\therefore$  as  $k$  exists as a complex number

then  $\frac{z}{w}$  exists as complex number

hence complex numbers  $z$  and  $w$  exist

such that  $\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w}$

eg  $z = -1 + 3i$  and  $w = 2$

3 marks – correct solution

2 marks – simplifies the question by assigning a value for either  $z$  or  $w$

1 mark – correct solution for assumption that  $|z| = |w| = 1$

|     |                                                                                                                                                                                                                                               |        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| d-i | $a^2 + b^2 + c^2 \geq ab + ac + bc$ $(a - b)^2 \geq 0$ $a^2 - 2ab + b^2 \geq 0$ $a^2 + b^2 \geq 2ab$ <p>Similarly</p> $a^2 + c^2 \geq 2ac$ $b^2 + c^2 \geq 2bc$ $2a^2 + 2b^2 + 2c^2 \geq 2ab + 2ac + 2bc$ $a^2 + b^2 + c^2 \geq ab + ac + bc$ | 1 mark |
| e-i | $(a + b + c)^2 \geq 3(ab + ac + bc)$ $\text{LHS} = (a + b + c)^2$ $= a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$ $= a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$ $\geq ab + bc + ac + 2ab + 2bc + 2ac$ $\geq 3ab + 3bc + 3ac$ $\geq 3(ab + bc + ac)$             | 1 mark |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| e-ii | <p><i>hence prove: <math>a^2 + b^2 + c^2 \geq 3\sqrt[3]{a^2b^2c^2}</math></i></p> $(a - b)^2 \geq 0$ $a^2 - 2ab + b^2 \geq 0$ $a^2 - ab + b^2 \geq ab$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 + b^3 \geq (a + b)ab \dots (1)$ <p><i>Similarly</i></p> $b^3 + c^3 \geq (c + b)cb \dots (2)$ $a^3 + c^3 \geq (a + c)ac \dots (3)$ <p><i>Multiply Lines 1, 2, 3 by <math>\frac{c}{c}, \frac{a}{a}</math> and <math>\frac{b}{b}</math> gives</i></p> $a^3 + b^3 \geq \left(\frac{a}{c} + \frac{b}{c}\right)abc$ $b^3 + c^3 \geq \left(\frac{b}{a} + \frac{c}{a}\right)abc$ $a^3 + c^3 \geq \left(\frac{a}{b} + \frac{c}{b}\right)abc$ <p><i>Now add these new inequalities</i></p> $2(a^3 + b^3 + c^3) \geq abc\left(\frac{a}{c} + \frac{b}{c} + \frac{b}{a} + \frac{c}{a} + \frac{a}{b} + \frac{c}{b}\right)$ $2(a^3 + b^3 + c^3) \geq abc\left(\frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b}\right)$ $2(a^3 + b^3 + c^3) \geq abc\left(\frac{a^2 + b^2}{ab} + \frac{a^2 + c^2}{ac} + \frac{b^2 + c^2}{bc}\right)$ $nb \ a^2 + b^2 \geq 2ab$ $\therefore 2(a^3 + b^3 + c) \geq abc\left(\frac{2ab}{ab} + \frac{2ac}{ac} + \frac{2bc}{bc}\right)$ $2(a^3 + b^3 + c^3) \geq 6abc$ $a^3 + b^3 + c^3 \geq 3abc$<br>$\text{Let } a \rightarrow \sqrt[3]{a} \quad b \rightarrow \sqrt[3]{b} \quad c \rightarrow \sqrt[3]{c}$ $a + b + c \geq 3\sqrt[3]{abc}$<br>$\text{Let } a \rightarrow \sqrt[3]{a^2} \quad b \rightarrow \sqrt[3]{b^2} \quad c \rightarrow \sqrt[3]{c^2}$ $a^2 + b^2 + c^2 \geq 3\sqrt[3]{a^2b^2c^2}$ | <p>3 marks - correct</p> <p>2 marks – correct however used AM v GM<br/>(Note: if “Hence” had not been used in the question – it would have been legitimate to use AMvGM statement.</p> <p>1 mark – some progress using part i and ii</p> |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Question 16

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a-i  | <p>Clearly, <math>z = -1 = \text{cis } \pi</math> is a solution.</p> <p>The solutions lie on the unit circle, equally spaced.</p> <p>Hence, solutions are:</p> $z = \text{cis } \pi, \text{cis } \frac{\pi}{3}, \text{cis } \left(-\frac{\pi}{3}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>2 marks – correct solution</p> <p>1 mark - Writes one of the solutions in modulus-argument form</p>                                                                                                                                                                     |
| a-ii | <p>Using part (ii),</p> $z^2 + 2 = \text{cis } \pi, \text{cis } \frac{\pi}{3}, \text{cis } \left(-\frac{\pi}{3}\right)$ <p><math>z^2 + 2 = \text{cis } \pi</math>:</p> $z^2 + 2 = -1$ $z^2 = -3$ $z = \pm i\sqrt{3}$ <p><math>z^2 + 2 = \text{cis } \frac{\pi}{3}</math>:</p> $z^2 + 2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ $z^2 = -\frac{3}{2} + \frac{\sqrt{3}}{2}i$ $z^2 = \sqrt{3} \text{cis } \frac{5\pi}{6}$ $z = \sqrt[4]{3} \text{cis } \frac{5\pi}{12}, \sqrt[4]{3} \text{cis } \left(\frac{5\pi}{12} - \pi\right)$ $z = \sqrt[4]{3} \text{cis } \frac{5\pi}{12}, \sqrt[4]{3} \text{cis } \left(-\frac{7\pi}{12}\right)$ <p><math>z^2 + 2 = \text{cis } \left(-\frac{\pi}{3}\right)</math>:</p> $z^2 + 2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ $z^2 = -\frac{3}{2} - \frac{\sqrt{3}}{2}i$ $z^2 = \sqrt{3} \text{cis } \left(-\frac{5\pi}{6}\right)$ $z = \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12}\right), \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12} + \pi\right)$ $z = \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12}\right), \sqrt[4]{3} \text{cis } \left(\frac{7\pi}{12}\right)$ <p><math>\therefore</math> The six solutions are:</p> $z = \pm i\sqrt{3}, \sqrt[4]{3} \text{cis } \frac{5\pi}{12}, \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12}\right), \sqrt[4]{3} \text{cis } \frac{7\pi}{12}, \sqrt[4]{3} \text{cis } \left(-\frac{7\pi}{12}\right)$ | <p>3 marks – correct solution</p> <p>2 marks - Finds two solutions, each from a different substitution of the 3 expressions from (i), or equivalent merit.</p> <p>Equivalent forms of solutions were accepted.</p> <p>1 mark - Finds one solution, or equivalent merit</p> |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                       |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| b-i   | <p>Find <math>\overrightarrow{PR}</math> and <math>\overrightarrow{QS}</math> in terms of <math>e</math>.</p> $\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP}$ $\overrightarrow{OR} = \frac{\overrightarrow{OE} + \overrightarrow{OC}}{2} = \frac{4e\mathbf{k} + 4\mathbf{i} + 4\mathbf{j}}{2} = 2\mathbf{i} + 2\mathbf{j} + 2e\mathbf{k}$ $\overrightarrow{OP} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2} = -4\mathbf{j}$ $\therefore \overrightarrow{PR} = 2\mathbf{i} + 6\mathbf{j} + 2e\mathbf{k}$ <p>Similarly,</p> $\overrightarrow{QS} = -6\mathbf{i} - 2\mathbf{j} + 2e\mathbf{k}$ | <p>2 Marks: Provides correct solution.<br/>(Must include working for QS)</p> <p>1 Mark: Finds <math>\overrightarrow{PR}</math> or <math>\overrightarrow{QS}</math>, or equivalent merit</p>                                                                                                           |
| b-ii  | <p>For Line <math>l</math>, use point <math>P</math> and direction vector <math>\overrightarrow{PR}</math>:</p> $\therefore \text{Line } l: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ 2e \end{pmatrix}$ <p>For Line <math>s</math>, use point <math>Q</math> and direction vector <math>\overrightarrow{QS}</math>:</p> $\therefore \text{Line } s: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -6 \\ -2 \\ 2e \end{pmatrix}$                                         | <p>2 Marks: Provides correct solution.<br/>OR<br/>Obtains equations consistent with an error in part (i)</p> <p>1Mark: Writes the vector equation of either Line <math>l</math> or Line <math>s</math>, or equivalent merit<br/>OR<br/>Obtains the correct direction vector for <b>each</b> line.</p> |
| b-iii | <p>To find the point of intersection, equate the <math>x</math> components:</p> $2\lambda = 4 - 6\mu$ $\therefore \lambda = 2 - 3\mu \quad [1]$ <p>Equate the <math>y</math> components and use [1]:</p> $-4 + 6\lambda = -2\mu$ $-4 + 6(2 - 3\mu) = -2\mu$ $\therefore \mu = \frac{1}{2} \quad \text{and} \quad \lambda = \frac{1}{2}$ <p><math>\therefore</math> Point of intersection is <math>(1, -1, e)</math></p> $\therefore \overrightarrow{OM} = \mathbf{i} - \mathbf{j} + e\mathbf{k}$                                                                                                                       | <p>2 Marks - Uses the values of the parameters to find the point of intersection, making one error</p> <p>1 Mark - Correctly equates <math>x</math> and <math>y</math> components and attempts to solve simultaneously to find the values of the parameters</p>                                       |
| b-iv  | $\overrightarrow{EB} = \overrightarrow{OB} - \overrightarrow{OE} = 4\mathbf{i} - 4\mathbf{j} - 4e\mathbf{k}$ <p>If <math>\overrightarrow{OM} \perp \overrightarrow{EB}</math>, <math>\overrightarrow{OM} \cdot \overrightarrow{EB} = 0</math>:</p>                                                                                                                                                                                                                                                                                                                                                                     | <p>3 Marks: Correct answer</p> <p>2 Marks: Obtains two correct values for <math>e</math>, <math>\cos\theta</math> or <math>\theta</math></p>                                                                                                                                                          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{pmatrix} 1 \\ -1 \\ e \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -4 \\ -4e \end{pmatrix} = 0$ $4 + 4 - 4e^2 = 0$ $e = \sqrt{2} \quad (e > 0)$ <p>The angle between <math>\overrightarrow{PR}</math> and <math>\overrightarrow{QS}</math>:</p> $\overrightarrow{PR} = 2\mathbf{i} + 6\mathbf{j} + 2\sqrt{2}\mathbf{k}$ $\overrightarrow{QS} = -6\mathbf{i} - 2\mathbf{j} + 2\sqrt{2}\mathbf{k}$ $\therefore \cos \theta = \frac{\overrightarrow{PR} \cdot \overrightarrow{QS}}{ \overrightarrow{PR}   \overrightarrow{QS} } = \frac{-12 - 12 + 8}{\left( \sqrt{2^2 + 6^2 + (2\sqrt{2})^2} \right)^2} = -\frac{1}{3}$ $\therefore \theta \cong 109^\circ \text{ (nearest degree)}$ | <p>OR</p> <p>Obtains values of <math>\cos \theta</math> and <math>\theta</math> consistent with an incorrect value of <math>e</math>.</p> <p>1 Mark: Finds a correct expression for vector EB</p> <p>OR</p> <p>Finds the value of <math>e</math></p> <p>OR</p> <p>Finds an angle consistent with incorrect values for EB and <math>e</math>.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|